Review of User Experience and User Interface Design for Children's Educational Apps

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***Abstract*—A good user interface and user experience are the cornerstones for developing an application or website that can be used effectively. Naturally, students who are adjusting to the present pandemic situation need time to change their learning techniques, particularly if those approaches are entirely digital. The PRISMA approach, which included a thorough evaluation of the literature, was the principal tactic employed by the authors of this study. The PRISMA research methodology was chosen by the author because it permits reporting of the number of reviews and meta-analyses that were carried out while producing this study and also covers broad issue themes. For a website or program to be well-liked by users, UI/UX is well-known to be crucial. The appearance of a website or application without UI/UX will be uninteresting and confusing. The goal of this study is to identify a user interface and a user experience in children's educational applications.**

Keywords—User Experience, User Interface, Education, Learning, Interactive Virtual Training

I. Introduction

Schools are one of the public facilities that are unavailable because of the current pandemic. For kids, in particular, school is a place to learn. Online education is one of the current solutions offered by the government. Children must use a variety of online resources during this online learning, including Zoom, Google Meet, Google Classroom, and others [1].

All children, especially those in elementary school and below, have had to adapt because of the pandemic situation. These schoolchildren must be able to adapt to utilizing the online resources that have been made available [2], but at such a young age, it must be quite challenging for them to comprehend and swiftly adjust to the system. The UI/UX design of a system is one of the crucial components for a user to comprehend it [3].

The first step in making an application or website easy to understand and use is to create a decent user interface and user experience. Naturally, school pupils who are adjusting to the current pandemic condition need time to alter their learning styles, particularly with learning styles that are solely digital [4].

Online education is still regarded as less efficient. Moreover, learning programs UI/UX is not developed at all. Since many learning applications still have user interfaces and user experiences (UI/UX) that are not appropriate for school students, the author of this study suggests looking for UI/UX that is very appropriate for students so that they can quickly adapt to and understand the learning applications they will use [4].

We wish to identify a suitable UI/UX for children-friendly learning educational applications based on these issues for them to learn more effectively.

II. Method

A. Research Question

To assess the current state of research in the field of UI/UX Education. This literature review will answer two research questions

RQ1: Are a User Interface and User Experience suitable for educational learning applications?

RQ2: How big is the impact or UI/UX effect that affects a learning system application?

RQ1 focuses on finding a UI/UX design for several existing applications, ranging from educational learning applications to non-educational learning applications. Meanwhile, RQ2 focuses on identifying important aspects of UI/UX in an educational learning application system.

B. Search Strategy

A systematic search of the literature was carried out on the academic search engine Scopus. For the searched subject, IEEE, Research Gate, and Emerald databases are the most relevant to research needs.

The search focused on User Experience and User Interface, as supported by research on education, education, and teaching and learning. For the year of publication, we focus on between 2017 and 2022

C. Inclusion and Exclusion Criteria

Articles with the following characteristics, published between January 2017 and June 2022, with inclusion,

1. Paper is written in English
2. Talking about UI/UX for kids
3. Discuss UI/UX for the education system
4. There should be at least discuss of UI UX

Meanwhile, articles with the following criteria will not be included or excluded

1. Paper is not written in English
2. There are duplicates
3. studies that are not verified by professionals

Those are the inclusion criteria and exclusion criteria for our systematic literature review.

D. Screening Process and Results

304 journals were discovered in the initial step of screening, which separated the journals based on irrelevant themes, duplicate journals, and journals that weren't in English or Indonesian. After the initial screening, 159 journals are received.

The journals are divided in the second screening according to the requirements and how pertinent the journal topic is to the journal that we will create. 41 journals from the second screening were obtained.

41 journals in total have been used. through the selection procedure. Examine references and journals that made it past the second round of screening to determine which ones can be used to supplement the content. There will be 44 journals utilized in all.

E. Data Collection

44 articles or journals met the requirements after the screening procedure. We learned the following information from the publications or journals:

1. The source and full reference.
2. Method of how to develop an effective UI/UX.
3. UI/UX design's impact on kids' education.

F. Data Analysis

By locating the journal articles related to the study question, the collected data is de-identified. To make the comparison process simple, the contents of those journals are classified into tables. Concept tables go from the author's review of the literature to becoming concept-centric, giving readers help and structure for understanding concepts.

The primary focus of our review of the journals' content was on how to create an effective UI/UX for children's education (addressing RQ1) and how UI/UX design affects children's education (addressing RQ2).

When we cross-checked the journals' content against the study's focus, we discovered explanations on how to spot effective UI/UX design and how UI/UX design impacts education.

III. RESULT

These views let users relate to and interact with a product. The user interface not only acts as a link but also enhances aesthetics to make users happier. In addition to being visually appealing, the UI needs to be easy to use. A few examples of UI components include button elements, typographic icons, themes, layouts, product-specific animations, and other interactive visuals. All of these UI components were developed with an eye on both usability and aesthetics. Specifically, player context, game flow, immersion, game system, and learnability. Additionally, it should be mentioned that an image's arrangement, proper photos, hierarchy, visuals, and suitable color scheme make it appear appealing and not monotonous. The relevant publications are shown in Table I.

TABLE I. TABLE RELEVANT PUBLICATION

Electronic Database	Paper Found	Candidate Considerate	Publication Relevant to RQ
Google Scholar	104	18	16
Research gate	96	10	9
Scopus	54	8	7
Other	50	6	6
Total	304	44	38
Google Scholar	104	18	16

A. RQ 1: Are a User Interface and User Experience suitable for educational learning applications?

The response to RQ 1 is "yes". The result will be knowing which UI/UX is appropriate for schooling. Later, develop a successful alliance. In other words, the methodologies used in this program or website for evaluation and learning are how the educational results are determined. Because the more clearly stated the learning outcomes are, the easier it is to ensure positive alignment. As a result, three distinct base classes are created. Additionally, it should be mentioned that an image's arrangement, proper photos, hierarchy, visuals, and suitable color scheme make it appear appealing and not monotonous. The relevant publications for RQ1 are shown in Table II.

That's accurate when it comes to instructional games. To mention a few, there is fluidity, immersion, usability, gaming system, player context, and learnability [5]. Additionally, it should be mentioned that an image's arrangement, proper photos, hierarchy, visuals, and suitable color scheme make it appear appealing and not monotonous.

The Alomari [19] framework outlines the fundamental ideas of cyberlearning as well as the crucial characteristics connected to each idea. As in the example, one of the five key ideas in building and assessing a cyberlearning environment is technology. This technology was chosen for SEP-CyLE to use to conduct its review. UI/UX evaluation framework concepts and criteria for a cyberlearning environment are shown in Fig. 1. The five main concepts of the evaluation model are as follows:

1. Context: Used to assess the environment for online learning. It matters where the context is employed. SEP-CyLE, for instance, was created to facilitate students' understanding of software development principles, methods, and tools.
2. Users: As long as there are users who respect cyberlearning technology and services so that they have a purpose in their presence, the cyberlearning environment is highly important. Students and teachers are two different groups of users in SEP-CyLE. These two categories of users are included in the review process by Alomori [19], with students serving as the primary source of data.
3. Technology: Alomori [19] researches how people use technology to learn and how it may help them learn. Then, associated technologies utilized in the cyberlearning environment, such as gamification and collaborative learning, are collected along with the data needed for evaluation.
4. Utility: Alomori [19] examined the information gathered that is connected to the technology itself to assess the technology's utility. The analysis procedure takes into account the user's demands and pleasure when using that technology to comprehend the given environment.
5. Usability: Alomori [19] measures how simple it is for a particular type of user to use a particular technology in a particular setting to promote learning. For instance, they measured the subjective and objective values of group learning and problem-based (task-based) learning, respectively, using heuristic evaluation and cognitive walkthroughs.

TABLE II. RELEVANT PUBLICATION TO RQ1

Technique Used	Number Of Papers	Paper Reference
User Interface	8	[6],[7],[8],[9]
User Experience	16	[10],[11],[12]
Mobile Application	5	[13],[14],[15],[16],[17]
Educational Learning	15	[18],[19],[20],[21]

B. RQ 2: How big is the impact or UI/UX effect that affects a learning system application?

For RQ 2 answer, currently, the role of UI/UX is known and is known to be very important for a web or application to be liked by users. Without UI/UX the appearance of an application or web will be boring and unclear. a web or application requires an interactive UI/UX design and will surely be liked by many users. Besides that, to be used in learning, UI/UX must look at the age range, whether it is for children from kindergarten to elementary school or for junior and senior high school students, of course, the UI/UX provided will also be different in each age range or class level. Another benefit is to be able to help teachers who teach because technology is growing here and it is also growing and children should be taught technology from an early age. Therefore, it can be concluded that UI/UX is very important for a learning application [4]. The relevant publication for RQ2 is shown in Table III.

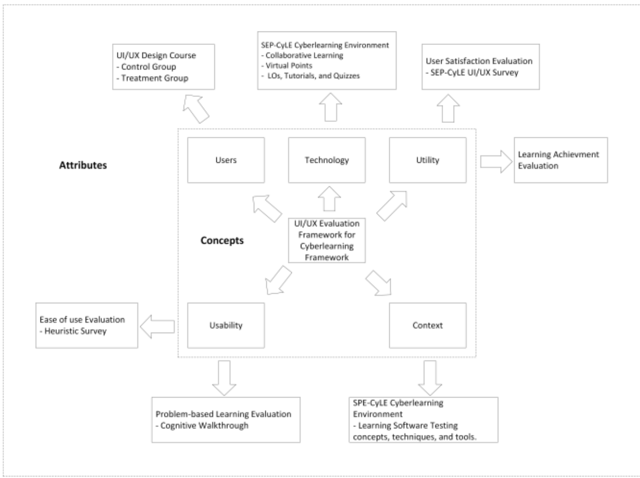

Fig. 1 UI/UX evaluation framework concepts and criteria for a cyberlearning environment [19].

TABLE III. RELEVANT PUBLICATION TO RQ2

Technique Used	Number Of Papers	Paper Reference
User Interface	8	[6],[7],[8],[9]
User Experience	16	[10],[11],[12]
Learning System	5	[13],[14],[15],[16],[17]
Educational Learning	15	[18],[19],[20],[21]

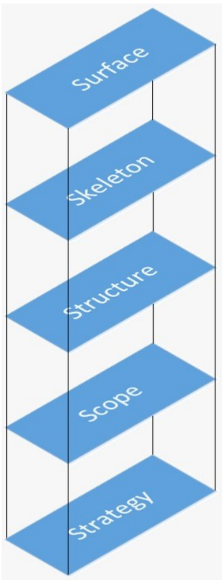

Fig. 2 Five planes conceptual framework for user experience [13].

Putra [13] was concerned about the large number of sites that are difficult to use and unfriendly to users because of which users are reluctant to use the site and go to another site. The author claims that it is crucial to carry out research to evaluate UX in educational information systems because information technology has been used in various industries, including education. The findings indicate that the educational system has changed, both for teaching and for learning. Further research is required to lessen the problems users experience in understanding the system's goal as a result of the shifting process. An information system that users of the system can easily understand is the right information system. To entice users to use the website, website developers must be able to incorporate a number of components, as depicted in Fig. 2. The surface plane, skeleton plane, structure plane, and strategy plane elements make up this group.

IV. Discussion

Nowadays, learning is done both offline and online at the same time. E-learning programs are being used more frequently as a result, and building an online environment is becoming more popular. It promotes students' thorough preparation through a mix of traditional and online study. Based on the research, the article's author investigated the UI/UX design of a learning application or website. When creating user interfaces, it's crucial to consider the layout, visuals, visual hierarchy, color schemes, typography, readability, adaptability, navigation, and content. We hypothesize that students' interest in and comfort level with studying through applications or websites would be influenced by the user interface and user experience design. To determine which UX should be used in online learning applications and websites, this research also examines a variety of user experiences.

A. UI/UX

The process of developing a product with the consumer in mind is known as user experience (UX). You can make things utilizing this strategy that users' wants and preferences. Customers should have a user-friendly experience when using your product, according to UX design. Users of the product experience comfort and convenience when using it. The visual facets of design and all user-interaction-related features are included, and this UX component also covers how the product's components are arranged and navigated for use. UX also covers the definition of your branding, content, and copywriting that is acceptable for your target audience. UI is a part of UX because it provides a visual representation of a system's design. These views let users relate to and interact with a product. The user interface not only acts as a link but also enhances aesthetics to make users happier. In addition to being visually appealing, the UI needs to be easy to use. A few examples of UI components include button elements, typographic icons, themes, layouts, product-specific animations, and other interactive visuals. All of these UI components were developed with an eye on both usability and aesthetics.

In a study by J. Hoensik et al. [22], the fundamentals of UI/UX were broken down into 18 lists, which were then grouped into 4 categories, and assessed on 29 students in the IT department. As a result, only 20% of pupils comprehend UI/UX while 80% do not. Additionally, D. P. Kristiadi et al. [23] examined the effects of UI, UX, and GX on the video game Tekken 7. Only 52 of the study's 60 participants, according to the author, had ever played Tekken 7. The findings revealed that Tekken 7 had a fantastic UI, UX, and GX and that the typical participant was a guy between the ages of 21 and 30. To enhance an AR-HUD (Augmented Reality-Head Up Display) on ADAS, M.K. Choi et al. [24] develop a simulation framework employing an inference model based on a region-based fully convolutional network (Autonomous Driver Assistant System). The goal of the research is to reduce display error as seen by the driver. A new, 3D-based user interface (UI) was created for the Bomberman game by N. Zulfa et al. [25] under the name Bomba. The GEQ (Game Experience Questionnaire) method was used to test the game. The findings revealed that 62 users thought the game's UI and UX were substantially better than in the original Bomberman game.

B. UI/UX for Education

How user-friendly the User Interface (UI) and User Experience (UX) depend on the structure of the framework plan, which separates these two tasks into individual interactions to build a school system. By convention, the term "Client Experience" (UX) refers to knowledge and human responses resulting from the anticipated use or utilization of a product, framework, or administration. Client insight includes thoughts, habits, coping mechanisms, actions, and successes that take place before, during, and following usage. The brand image, functionality, usability, effectiveness of the framework, intuitive method of acting, and additional boundaries of frameworks, products, or administration all have an impact on customer communication.

It is also a byproduct of the client's mental and emotional state, resulting from interactions, thinking, a person's past skills, and goal-setting. A more comprehensive interpretation of this term is discussed in certain literature. Despite the increased interest in framework usage experiences, it is challenging to reach a consensus on the type and scope of UX usage. Most study participants concurred that user experience (UX) is dynamic and dependent on settings and emotions, including factors like time, location, and reason. The developers propose defining UX as something personal (unfriendly) that arises through interaction with products, frameworks, services, or texts in answer to increasingly hostile questions.

However, to encompass everything that can affect the client experience, the phrases utilized need to be further defined. UX chooses in-depth, targeted elements for particular interactions and fosters faith in things. The inner state of the client (tendencies, assumptions, needs, inspiration, temperament, and so on), elements of the planned framework (e.g., complexity, reasoning, ease of use, usability, etc.), and communication (e.g., authoritative climate/social, movement content, through and through freedom to use, and so on) all have an impact on user experience (UX). An optimal client experience (UX) must first and foremost satisfy the customer's unique requirements while avoiding conflict and annoyance. Then create goods that are enjoyable to own and utilize by acting honestly and stylishly. A sincere client experience goes beyond the client's needs, wants, or the fundamental goal of the agenda. It's critical to continually integrate administration across multiple areas, plans, exhibits, visual computerization, and industry, including connection point plans, to achieve UX excellence in organizational contribution.

C. Teacher-Student-Computer Interaction.

Children are increasingly using cell phones in their daily lives. Therefore, it is crucial to properly boost their applications for them. Sequences for Kids and Patterns, an application with a straightforward design that kids can undoubtedly complete, is terrific software for kids in the learning process. Now, however, the focus will be on PC-based student-to-student communication.

Three components—teachers, students, and computers—are already present in web-based education, educational experiences, and virtual meetings. Sincere to goodness, PC serves as a point of contact between the instructor and pupils, who take on the roles of learners or recipients of learning. Executive class proficiency is necessary for a web-based learning strategy like this. The option for the instructor to strengthen the executive ability study room should be available in this circumstance. The Interactive Virtual Training for Teachers (IVT-T) framework can help educators deal more effectively with their homeroom teachers. IVT-T is a framework that can help teachers improve their abilities in their study rooms.

Usually, educators put forth an effort and make mistakes in homeroom teachers to hone their management skills, resulting in unfavorable growth opportunities, awkward homeroom settings, and stressful relationships. Nowadays, educators rely more on digital texts than on textbooks. mainly because it is usually easily gained by the kids and hence deserved. Teachers were in favor of digital books as long as they allowed their students to finish more volumes and made the teachers' jobs easier.

V. CONCLUSION

We found that children's applications feature robust user interfaces and user experiences that are highly accurate and harmonic, suggesting that the educational application's procedures and evaluation strategy are aligned with the learning outcomes. Because the more clearly stated the learning outcomes are, the easier it is to generate positive alignment. To make an image interesting and not boring, UI/UX must also pay attention to the image's structure, selecting the appropriate picture, hierarchy, graphics, and color scheme.

An application's UI/UX, particularly one that is used for learning and education, has a significant impact. We are all aware that when users start an application for the first time, there is a learning curve or, more accurately, an adaptation process that takes place. The application's UI/UX, of course, has an impact on this adaptation process. Because of this, UI/UX has a significant influence on instructional applications for children.

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